

14.5.4 Physical properties

1. Melting and boiling points

The boiling points of monohalo benzene which are all liquids follow the order

Iodo > Bromo > Chloro

The boiling points of isomeric dihalobenzene are nearly the same

The melting point of para isomer is generally higher than the melting points of ortho and meta isomers. The higher melting point of p-isomer is due to its symmetry which leads to more close packing of its molecules in the crystal lattice and consequently strong intermolecular attractive force which requires more energy for melting

p –Dihalo benzene > o- Dichloro benzene > m-Dichloro benzene

2. Solubility

Haloarenes are insoluble in water because they cannot form hydrogen bonds with water ,but are soluble in organic solvents

3. Density

Halo arenes are all heavier than water and their densities follow the order.

Iodo benzene > Bromo benzene > Chloro benzene

14.5.5 Chemical properties

A. Reactions invoving halogen atom

1. Aromatic nucleophilic substitution reaction

Halo arenes do not undergo nucleophilic substitution reaction readily. This is due to C–X bond in aryl halide is short and strong and also the aromatic ring is a centre of high electron density.

The halogen of haloarenes can be substituted by OH^- , NH_2^-, or CN^- with appropriate nucleophilic reagents at high temperature and pressure.

For Example

$$\underset{\text{Chlorobenzene}}{C_6H_5Cl} + NaOH \xrightarrow[300\ atm]{350^oC} \underset{\text{Pheno1}}{C_6H_5OH} + NaCl$$

This reaction is known as Dow's Process

$$\underset{\text{Chlorobenzene}}{C_6H_5Cl} + 2NH_3 \xrightarrow[50\ atm]{250^0C} \underset{\text{Aniline}}{C_6H_5NH_2} + NH_4Cl$$

$$\underset{\text{Chlorobenzene}}{C_6H_5Cl} + CuCN \xrightarrow[\text{Pyridine}]{250^0C} \underset{\text{Phenyl cyanide}}{C_6H_5CN} + CuCl$$

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